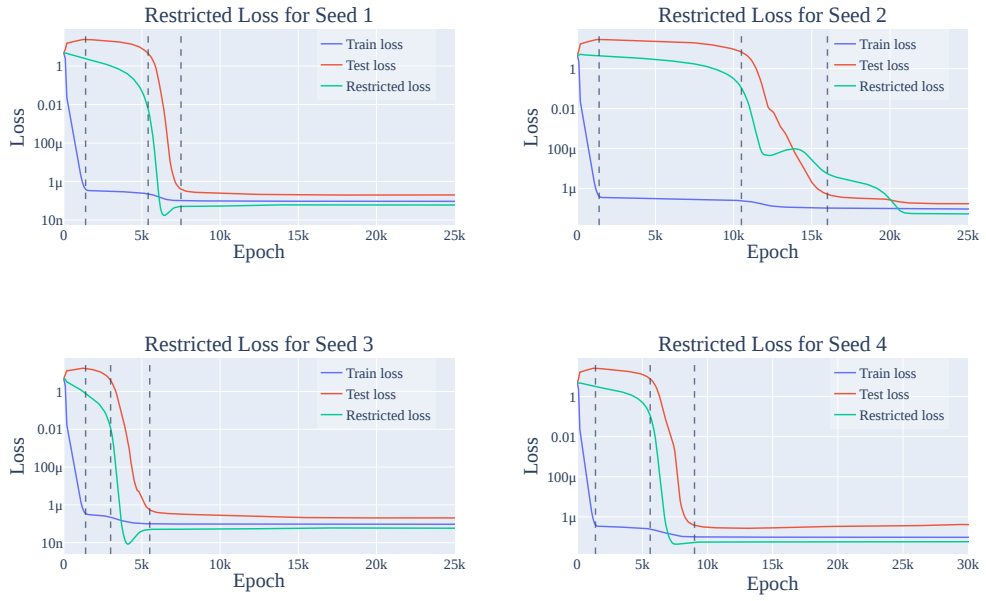

Figure 18: The train, test, and restricted loss for each of the four other random seeds described in Appendix C.2.1. The lines delineate the 3 phases of training: memorization, circuit formation, and cleanup (and a final stable phase). As with the mainline model, restricted loss consistently declines prior to train loss. Note that while the shapes of the loss curves are similar to each other and those of the mainline model, the exact time that grokking occurs (and thus the dividers between the phases of grokking) differ by random seed. Interestingly, memorization is complete by around 1400 steps for all five runs.

identified in the main text. Interestingly, while all of the models complete memorization by around 1400 epochs, circuit formation and cleanup occur at different times.

C.2.2 Results for other experimental setups

In this section, we provide further evidence that small transformers grok on the modular addition task, by varying the size of the network, the amount of training data, and the size of the prime P.

1-Layer Transformers with Varying Fractions of Training Data. We find that grokking occurs for the modular addition task with $P = 113$ for many data fractions (that is, the fraction of the $113 \cdot 113$ pairs of inputs that the model sees during training), as shown in Figure 20. Smaller amount lead to slower grokking, but sufficiently large fractions of data ($\geq 60\%$) lead to immediate generalization, as shown in Figures 20 and 21.

As with the results in Appendix C.2.1, all of the 1-layer transformers in this section also converge to using the Fourier multiplication algorithm.

2-Layer Transformers. As shown in Figure 22, 2-layer transformers also exhibit some degree of grokking. However, this is complicated by the slingshot mechanism (Thilak et al., 2022). We display the excluded loss of a 2-layer transformer in Figure 23 and find it shows a similar pattern to the mainline 1-layer transformer, in that it improves relatively smoothly *before* grokking occurs.

Smaller and larger primes. We also examined smaller and larger prime moduli. For $P = 53$ (Figure 24), we explored a variety of weight decays to observe grokking in the small prime case. With the original weight decay setting of $\lambda = 1$, we found that the models never generalized. However, increasing the weight decay to $\lambda = 5$ does allow the model to grok. We speculate that

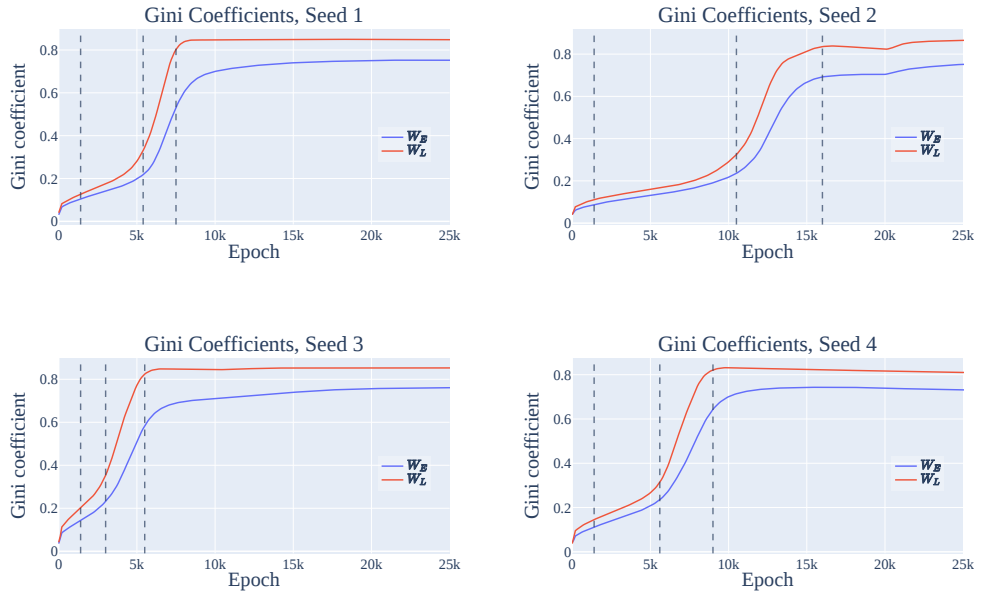

Figure 19: The Gini coefficients (a measure of sparsity) of the Fourier components of the embedding matrix W_E and the neuron-logit map W_L for each of the four other random seeds. The lines delineate the 3 phases of training: memorization, circuit formation, and cleanup (and a final stable phase). As with the mainline model, sparsity increases slowly during memorization and circuit formation, and then quickly during cleanup.

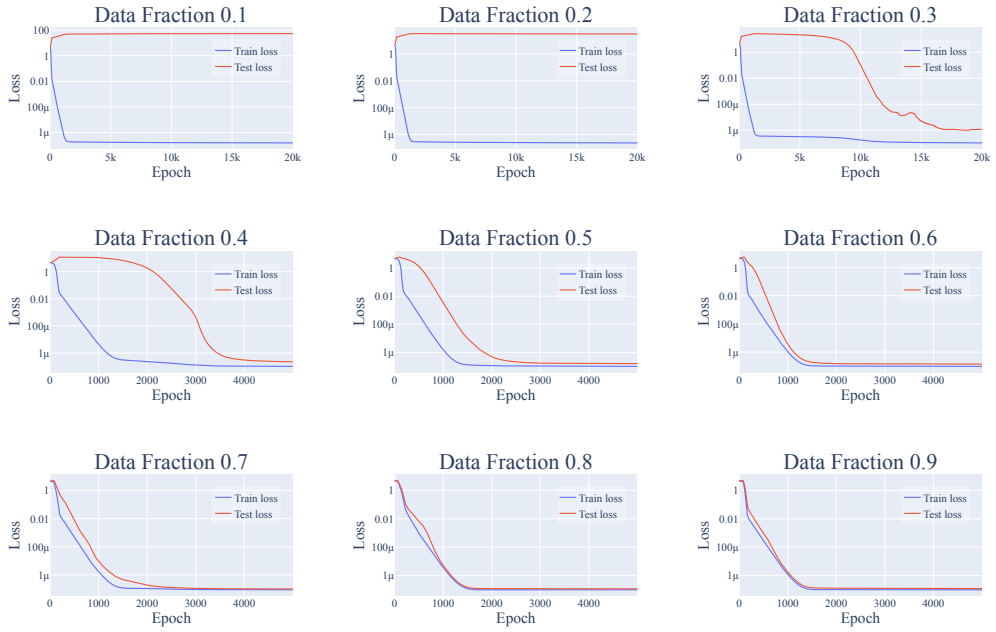

Figure 20: Training and test losses for a 1-layer transformer on the modular addition task with $P = 113$, with varying fractions of the $113 \cdot 113$ pairs of possible inputs used in training. Grokking occurs when between $30 - 50\%$ of the dataset is used during training and lower fractions of data lead to slower grokking. Using $\geq 60\%$ data leads to immediate generalization, while using 10% or 20% of the data doesn't lead to grokking even after 40k epochs. Note the different x-axes: we only show 5k epochs for the runs with data fraction $\geq 40\%$ for more detail.

Figure 21: Number of steps for train/test loss to be $< 10^{-6}$, as a function of the amount of training data. While train loss immediately converges to below 10^{-6} for all data fractions, generalization takes significantly longer with lower fractions of data. Note that the plots for other thresholds are also qualitatively similar.

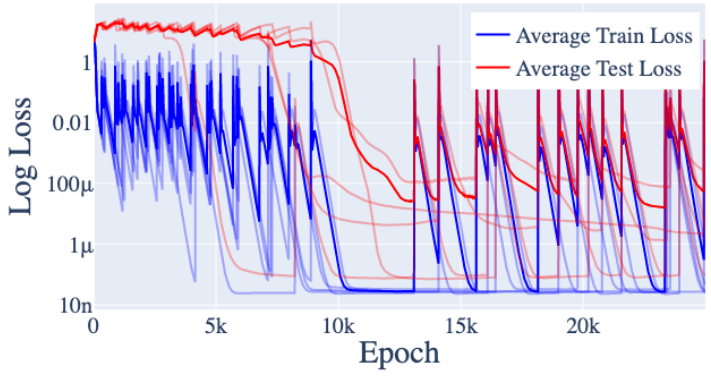

Figure 22: Training and test loss for a 2-layer version of the original architecture. Average across 5 random seeds is in bold.